UNIT42 · FIRMWAREX

Firmware Security Assessment

ARM32 controlled E2E firmware

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Executive summary

This assessment recorded **16 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	9	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	ARM32	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	ARM32	Present	Not detected	partial	Not detected	Not detected

Detailed findings

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1. Command injection candidate High

Property	Result
CWE	CWE-78
Affected file	/bin/taint-demo
Architecture	ARM32
Confidence	probable
Review status	new
Address	0x100004
Function / sink	system
Detected	2026-10-20T05:30:32.833586Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x10450 → 0x1034c → 0x10000c → 0x10478 → 0x10490 → 0x10328 → 0x100004
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_network_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_network_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args … [excerpt; full expression in JSON export]
stack hash	e360df36f4e1c32877e7587a1d348a8ffb66754f4e78a34b88a5ec61a38c98af

Evidence reference: evidence:20dc32c1dcd9e15242505c4aadd403bc395939faaf25e20445f1c2337350fa23. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review whether external input can reach command execution. Prefer structured APIs and strict allowlists; validate the candidate before remediation.

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2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923002Z

Evidence

Observation	Value
rule	credential
fingerprint	15e3f926eb962a1b96856d42ceb0d97bcc8e35061c586cdef9562889dfe76689
byte offset	9
byte length	19

Evidence reference: evidence:9f690ee6358a7dfc78ca5899c219eed1266b76f97ebac9e0a8685775bb3ee918. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923038Z

Evidence

Observation	Value
rule	credential

Observation	Value
fingerprint	f266d519df16fdc2fe4b518529a288cf101fa9474ea44aa8fadea13df6206f39
byte offset	37
byte length	24

Evidence reference: evidence:390f4cc992505f9a0f2a72ab809d9f26977ccc8b5bd25ba942ba49fc4649800d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923059Z

Evidence

Observation	Value
rule	private_key
fingerprint	04812bdf821a8ed85428211740fde078b847a80b827a642400004020c41053fa

Observation	Value
byte offset	62
byte length	78

Evidence reference: evidence:41c922c65889f91cd9e60792d7a9f2fcc288a15488fd8106c7ebf3d00f076afb. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923139Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	0e19ab164d2e94ffe6a19747b35916a5f7d6f8f312ec4bf4880dcbda29972499
byte offset	237
byte length	3

Evidence reference: evidence:0540884f422315d82c17696782c5e1e059f93d5a6a57e5c81bfd48d1e62970f7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923156Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	61521ee44a5153b011e595e6480a6eae2a7c9c3f396f1e7e06dacf0121828b3c
byte offset	255
byte length	3

Evidence reference: evidence:a2fa2dffc217b2e8eaa8274bdf430ecbcc181a0acbbdc4fef2025a254d0b47d7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-018cde065dc02e82
Architecture	ARM32
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923421Z

Evidence

Observation	Value
rule	credential
fingerprint	2d227a8956bb025a8155aff58e9570273f367d65b4c7e85795b6a34ccc0afbfb
byte offset	38
byte length	25

Evidence reference: evidence:6c53145cf9f1185a68540c8db1dad4c0776c5de4e4d8b40f4e66b39062d3ffd4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:29.354511Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.922093Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	44ac217b6ead19c5d317843783f5c164b8b2d0ae6978430016b9b2580bb014cf
byte offset	400862
byte length	50

Evidence reference: evidence:cc5718a143d38c65922c5cac3c663bfe6fce2d478e40cd6aad88387b1c648e6c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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10. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM32

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.922162Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	8e8d132acecaf5c8eafeeaaa92fdc1bc08ce9108b99e55e8006ccc229a6ef98c
byte offset	476053
byte length	26

Evidence reference: evidence:1adf43e1d8b1a792265cdaf351d7649fcdfbe62d3c533a52d311dc9218ffa349. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM32
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.922188Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	2714542c978317661bffeecf83b47b07a077f0bcfbda8da2b71fc57f7f6c4ce8
byte offset	476085
byte length	26

Evidence reference: evidence:09468f586c7155c5ad553f75d2dc6be31859bd6e94829a5a547ee3a144416f67. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-20T05:30:32.922209Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	d5f3101d59502441ea21d9d88dda55e51b978bb230ec02e03c3cf80c0c3dca14
byte offset	476944
byte length	36

Evidence reference: evidence:ea51de8148001897ac331ef3dbd5b8c8b97d49ca4218a90fca94079e275904e1. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.922229Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	9acbbc3f1279843ddb13b529e63776d2600d8d17d4db8c95d46ae0d05093b757
byte offset	476984
byte length	36

Evidence reference: evidence:d074987671124b5dd484c92a8834a80ebfd60b3064e0e87fe898e8e178f0dd26. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923084Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	aa57c6402ff7ac03b3cf1be0fbd3d0105de83a03fa0049c38850d529cb8bf501
byte offset	141
byte length	75

Evidence reference: evidence:83bde4f932592497fbe8d65e8446f9973921c2a8cc65c4b76aae1fb460912171. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923103Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	68aeb441a2c166969840c5887147daa30a56a1c921ab0a9ab2cf80a43616bae9

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:db9c25d2d97a11590842ad2e76b90aed366c9bc071637b9d1371cf1dd069f8e1. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	ARM32
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:32.923121Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	46a4dbc599bd270174b3f23a0ee7cf52f58ee4e3922b270459fc4e1549403c86
byte offset	259
byte length	29

Evidence reference: evidence:011d813cd7deb5304d08b430d7e3acbd03e4d1c61b46d54cf5c554fd9e192afa. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	16122d04-01f5-4376-a271-49d243b68d8f
SHA-256	974c196dd678242ab3b6e1ad0a79e48225b9e83244cc122eebe4e86332e67bd3
Job ID	841bca18-4e8a-4f9c-9c3e-91ddefb7e21d
Image bytes	282624